Neutron Kinetics without Multiplication

NE 630 | Nuclear Reactor Theory

25 LESSON

Read: FNRP 5.1–5.2 through Eq. (5.8).

Students will be able to write down, solve, and explain each term of a first-order, ordinary differential equation that models the neutron flux, the neutron density, or the number of neutrons in a non-multiplying system as a function of time.

1 Revisiting the BIG PICTURE™

In the course introduction, we saw the following horror:

$$\begin{aligned}\frac{1}{v}\frac{\partial\phi}{\partial t} &- \nabla\cdot D(\mathbf{r},E)\nabla\phi(\mathbf{r},E,t) + \Sigma_t(\mathbf{r},E)\,\phi(\mathbf{r},E,t)\\ &= \int_0^\infty \Sigma_s(\mathbf{r},E'\to E)\,\phi(\mathbf{r},E',t)\,dE'\\ &+ \frac{\chi(E)}{k_{\text{eff}}}\int_0^\infty \bar{\nu}(\mathbf{r},E')\Sigma_f(\mathbf{r},E')\phi(\mathbf{r},E',t)\,dE' + S(\mathbf{r},E,t)\,,\end{aligned}$$

which is the time-dependent, multigroup diffusion equation.

As for steady-state problems, we'll have $S(\mathbf{r},E,t)$ for source-driven problems (with or without multiplication) and k_{eff} for source-free problems with multiplication. For time-dependent problems, k_{eff} can first be computed for a steady-state model and then used to enforce a steady-state, initial condition.

2 The "Point" Kinetics Equation

We'll still ignore space. We won't *ignore* energy, but instead integrate it away:

$$\begin{aligned}\int_0^\infty dE\Bigg\{ \frac{1}{v}\frac{\partial\phi}{\partial t} + \Sigma_t(E)\phi(E,t) &= \int_0^\infty \Sigma_s(E'\to E)\phi(E',t)dE'\\ &+ \frac{\chi(E)}{k_{\text{eff}}}\int_0^\infty \bar{\nu}(E')\Sigma_f(E')\phi(E',t)dE'\\ &+ S(E,t)\Bigg\}.\end{aligned}$$

If we assume that $\phi(E,t)=\phi(t)\varphi(E)$, then we can write:

$$\frac{d\phi}{dt}\times\int_0^\infty\left(\frac{\varphi(E)}{v(E)}\right)dE + \bar{\Sigma}_t\phi(t) = \bar{\Sigma}_s\phi(t) + \frac{1}{k_{\text{eff}}}\bar{\nu}\bar{\Sigma}_f\phi(t) + S(t)\,.$$

or

$$\boxed{\frac{1}{\bar{v}}\frac{d\phi}{dt} + \bar{\Sigma}_t\phi(t) = \bar{\Sigma}_s\phi(t) + \frac{1}{k_{\text{eff}}}\bar{\nu}\bar{\Sigma}_f\phi(t) + S(t)}\,, \qquad (25.1)$$

where

$$\bar{\Sigma}_x = \frac{\displaystyle\int_0^\infty \Sigma_x(E)\,\varphi(E)\,dE}{\displaystyle\int_0^\infty \varphi(E)\,dE} \quad \text{and} \quad \boxed{\bar{\nu} = \frac{\displaystyle\int_0^\infty \varphi(E)\,dE}{\displaystyle\int_0^\infty \frac{\varphi(E)}{v(E)}\,dE}}\,. \qquad (25.2)$$

In other words, we've assumed the spectrum is independent of time but the overall neutron population is not. This is an imperfect but reasonably accurate assumption for many systems. Often, we call this a *point* kinetics model because it ignores spatial dependence (or treats it as one "point") and treats the neutron population as a single, time-dependent quantity.

3 The Mean Neutron Speed

The mean speed $\bar{v}$ represents a spectrum-averaged speed, and its numerical values are worth quantifying over the thermal, intermediate, and fast energy ranges we've previously explored.

Example: Assuming classical physics (i.e., $E=(1/2)mv^2$) applies, derive $v(E)$ cm s^{-1}, where E is given in eV.

Ans. The neutron mass is $m_n = 939.655\ \text{MeV}/c^2$. Thus

$$\begin{aligned}v(E)/c = \beta(E) &= \sqrt{(2E\ \text{eV}\cdot 10^{-6}\text{MeV eV}^{-1})/(939.655\ \text{MeV}/c^2)}\\ &= 4.614\cdot 10^{-5}\sqrt{E/c^2}\,,\end{aligned}$$

and, with $c = 3\cdot 10^{10}$ cm s^{-1}, we have

$$\boxed{v(E) = 1.384\cdot 10^6\sqrt{E}\ \text{cm s}^{-1}}\,.$$

A helpful sanity check: $v(0.0253\ \text{eV}) \approx 220{,}000\ \text{cm s}^{-1} = 2200\ \text{m s}^{-1}$.

Example: Assuming $v(E) = a\sqrt{E} = 1.384\cdot 10^6\sqrt{E}$ cm s^{-1} (E in eV), compute the mean neutron speed if

(a) $\varphi(E) = Ee^{-E/kT}$, $10^{-3} < E < 1$ eV, $T = 294$ K, $k = 8.6173\cdot 10^{-5}$ eV K^{-1}.
(b) $\varphi(E) = 1/E$, $1 < E < 10^5$ eV
(c) $\varphi(E) = \chi(E)$ (as defined by FNRP 2.31), $10^5 < E < 10^7$ eV .
(d) $\varphi(E)$, piecewise as above, with $\phi_T = \phi_I = \phi_F$.

Ans. Either numerical or, with suitable approximations, analytical techniques can be used. Here, just the numerical results based on `scipy.integrate.quad` are given:

(a) $2.498\cdot 10^5$ cm s^{-1}
(b) $7.992\cdot 10^7$ cm s^{-1}
(c) $1.513\cdot 10^9$ cm s^{-1}
(d) $7.266\cdot 10^5$ cm s^{-1} (notice how heavily weighted thermal neutrons are!)

4 Kinetics Sans Multiplication

Set $\bar{\Sigma}_f = 0$ in (25.1) to obtain

$$\frac{1}{\bar{\nu}}\frac{d\phi}{dt} + \bar{\Sigma}_a \phi(t) = S(t)\,. \tag{25.3}$$

Example: Rewrite (25.3) in terms of the neutron density $n'''(t)$.

Ans. Recall $\phi = n'''\bar{v}$. Thus,

$$dn'''/dt + \bar{\Sigma}_a \bar{v} n'''(t) = S(t)\,, \qquad \text{(Like FNRP 5.2)}$$

which is equivalent to FNRP (5.2) both sides are multiplied by the system volume.

Example: Suppose that $S(t) = S_0\delta(t)$ and that $n'''(0) = 0$. Find $n'''(t)$.

Ans. $n'''(t) = S_0 e^{-\bar{\Sigma}_a \bar{v} t}$

5 The Neutron Lifetime

If we consider $n(t)$ (or $n'''(t)$) as just found, or, if $S_0 = 0$ and we let $n(0) = n_0$ such that

$$n(t) = n_0 e^{-\bar{\Sigma}_a \bar{v} t}\,,$$

one may immediately see a similarity to

$$I(x) = I_0 e^{-\Sigma x} \qquad \text{and} \qquad n(t) = n(0)e^{-\lambda t}\,.$$

All three represent *exponential attenuation* in which

- $\Sigma^{-1} = \tau$ is the mean(-free) path to collision
- $\lambda^{-1} = \bar{t}$ is the mean time to decay
- $\boxed{l_\infty = [\bar{\Sigma}_a \bar{v}]^{-1} = \bar{t}}$ is also a mean time, but here it is the time a neutron travels, on the average, before being absorbed (and, hence, "dying").

Example. What's l_∞ for thermal neutrons in room-temperature water? Assume $\bar{\sigma}_a^{H_2O} = 0.6$ [b].

Ans. $N = 6.022 \cdot 10^{23}/18$, so $\Sigma_a = N\,\bar{\sigma}_a \approx 2.0 \times 10^{-2}$ cm^{-1}. For thermal neutrons, $\bar{v} \approx 2.5 \times 10^5$ cm/s. Thus $l_\infty \approx 2.0 \times 10^{-4} = 200\ \mu$s. Compare to https://journals.aps.org/pr/abstract/10.1103/PhysRev.61.152.